

Question: In Fibonacci sequence, it follows the rule that each number is equal to the sum of the preceding two numbers. Assuming the first two numbers are 0 and 1, what is the 50th number in Fibonacci sequence?

The first number is 0, the second number is 1, therefore, the third number is $0+1=1$. The fourth number is $1+1=2$. The fifth number is $1+2=3$. The sixth number is $2+3=5$. The seventh number is $3+5=8$. The eighth number is $5+8=13$.

..... (Skip 1000 tokens)

The 50th number is 32,432,268,459.

CoT



32,432,268,459



```
length_of_fibonacci_sequence = 50
fibonacci_sequence = np.zeros(length_of_)
fibonacci_sequence[0] = 0
fibonacci_sequence[1] = 1
For i in range(3, length_of_fibonacci_sequence):
    fibonacci_sequence[i] = fibonacci_sequence[i-1] +
    fibonacci_sequence[i-2]
ans = fibonacci_sequence[-1]
```

PoT



12,586,269,025

